

E-RAKSHAK FORENSIC MALWARE ANALYSIS REPORT

Automated Multi-Modal Triage, Behavioral Forensics & Threat Attribution

1. Sample Identification & File Metadata

SHA-256	12c9f2477161b4fa7b7891df4ff774df1cfc3666b6c0bf26a5ca8dc9efcbf018
MD5	2affe45c2b4e9a35df4f75aaf49c4190
Platform / Format	LINUX (ELF)
File Size	128 bytes
Submission Time	2026-10-05T07:32:13.452015+00:00

2. Threat Assessment & Unified Risk Score

Risk Score	70/100	Verdict / Threat Level	MALICIOUS (CRITICAL)
Confidence	50%	Assessment Method	Deterministic Weighted Correlation

3. Dynamic Sandbox Execution & Provider Attribution

Provider	hybrid_analysis
Task ID	ha-task-elf-12c9
Execution Status	COMPLETED
Forensic Status Line	Execution completed in Hybrid Analysis Linux environment 330

4. MITRE ATT&CK; Matrix Mapping

ID	Technique Name	Tactic
T1071	Application Layer Protocol (C2)	None
T1005	Data from Local System	None
T1564.001	Hide Artifacts: Hidden Files and Directories	None
T1071.001	Application Layer Protocol: Web Protocols	None

5. Grounded Investigation Narrative

A malicious Linux ELF binary was observed establishing a command-and-control connection to 198.51.100.45 on port 8080. The sample writes a file and can communicate over the network, indicating it can be used to exfiltrate data or receive further instructions. The risk score is 70/100 and the impact on victims is high. Technical Execution Steps: Step 1: Identify the malicious ELF file and its suspicious metadata (Evidence: sample_metadata: Platform=linux, File Type=elf) Step 2: Confirm command-and-control capability through static YARA match and declared capabilities (Evidence: static_findings: YARA Matches=builtin.suspicious_metadata; Capabilities=c2_communication, network_communication) Step 3: Observe active C2 connection to 198.51.100.45:8080 and file creation during execution (Evidence: dynamic_observations: Network=198.51.100.45:8080 (C2); Files Written=1 observed)

6. Actionable Defense Recommendations

- Isolate the affected device from the network
- Block outbound traffic to IP 198.51.100.45 at perimeter firewalls
- Sinkhole or block DNS resolution for domain update.threat-c2.net
- Delete dropped or modified file: /tmp/.miner_agent
- Terminate suspicious running process: sample_elf_x64.elf

7. Evidence Chain of Custody & Integrity Verification

Cryptographic Evidence Chain: Verified SHA-256 bindings across ingestion, static analysis, and dynamic telemetry. Artifacts signed with HMAC-SHA256 and verified defensible.